

Question 2

Question 2 — how the system works, and what has actually been measured.

The question

How does it work, and what have you measured?

My answer

Let me start with the honest part. **I have not measured this in a real plant.** Every number I have comes from a working model I built and tested using published figures, values derived from them, and a small number of stated assumptions. The product is designed to make that distinction visible on every screen rather than hide it.

Here is what the system does.

It takes a day's order book — the dye lots, their shade depths, fabric weights and ship dates — along with the machine's water allowance. It then works out, for each possible way of running that day, what the consequences are: water used, salt entering the effluent, concentrated reject volume, evaporator heat, steam, carbon, cost and whether any delivery slips.

The link that matters is this one. Reject volume is not guessed. It comes from salt mass divided by the highest concentration the membranes can safely reach, which I model at 60,000 mg/L. That is a stated model parameter, not a claim that every membrane in Tirupur runs at exactly that limit. Reject volume then sets evaporator heat, which sets steam, coal and carbon.

For the five-lot example, it evaluates 480 candidate plans — every running order crossed with every process strategy — and picks the best plan that is actually **feasible**, not simply the one with the lowest water number. Firm ship dates are hard constraints. A plan cannot buy an environmental saving by becoming operationally unacceptable.

Two human gates sit in the workflow. A named person approves the plan, and a second gate controls wash-off release. If a sensor is unreliable, the system holds rather than releasing, and credits exactly zero saving.

I tested it three ways.

First, 143 automated tests covering the calculations, the accounting invariants, the safety interlocks, record integrity, fault behaviour and concurrency.

Second, I checked the model against published plant data. The Central Pollution Control Board measured 18,340 mg/L of dissolved solids entering evaporation at a Tirupur unit. Separately, operators report reject running at 20–30 percent of inlet volume. Those two facts come from different places, and neither is an input to my engine. When I feed it the first, it predicts 30.6 percent — inside the reported band. At 12,000 mg/L it gives exactly 20.0 percent and at 15,000 it gives 25.0. Nothing was tuned to make that happen. This is a sanity check against published operating data, not a field validation.

Third, I ran a worked example on one machine, one shift, five lots. The baseline plan draws 11,907 litres of fresh water. The recommended plan draws 11,187 and cuts evaporator heat from 2,239 to 2,104 kWh. The lowest-water plan available uses only 5,889 litres — and the system refuses it, because it makes one delivery 2.6 hours late.

Of the 25 inputs driving the model, 17 come from published sources, 4 are derived from those, and 4 are openly labelled assumptions. **None are plant measurements yet.** The two measurements I need first are evaporator steam flow and reject conductivity.

Supporting notes

Why I answered it this way

The question asks what I measured, and the truthful answer is "nothing, in a plant." Any attempt to blur that would collapse the first time a judge asked a direct follow-up. Saying it in the opening line removes the risk entirely and buys credibility for everything after it.

The three-part test structure exists because "I tested it" means nothing on its own. Automated tests prove internal consistency. The CPCB comparison proves the physics behaves like real plants. The worked example proves it produces a decision, not just numbers. They answer three different doubts.

The refused plan is in the answer deliberately. Showing the system rejecting its own best-looking water result is the strongest available evidence that it is built for a factory rather than for a scoreboard.

Every figure, and how to verify it

Claim	Value	How to check
Automated tests	143	<code>python -m pytest tests/ -q</code>
Candidate plans evaluated	480	<code>figures.json</code> → candidates; 120 orders × 4 strategies
Membrane concentration ceiling	60,000 mg/L	<code>core/factors.py</code> → <code>ro_max_reject_tds_mg_l</code> (PUBLISHED, 15,000–80,000 band)
CPCB measured inlet	18,340 mg/L	CPCB Tirupur ZLD assessment
Model prediction at that inlet	30.6%	<code>tests/test_published_validation.py</code>
At 12,000 / 15,000 mg/L	20.0% / 25.0%	same test
Published reject band	20–30% of inlet	Indian ZLD operator data
Baseline freshwater	11,906.8 L	<code>figures.json</code> → <code>options.OPTION_A.fresh</code>
Recommended plan freshwater	11,186.8 L	<code>options.OPTION_B.fresh</code>
Evaporator heat, baseline → recommended	2,239.4 → 2,104.0 kWh	<code>options.OPTION_A/B.mee</code>
Lowest-water plan	5,888.8 L, 2.6 h late, refused	<code>options.OPTION_C</code>
Evidence mix	17 published / 4 derived / 4 assumed / 0 measured	<code>core/factors.py</code> ; <code>/api/evidence</code>

What I deliberately did not claim

- I did not call the CPCB comparison a validation of the product. It validates the *physics*, at conditions someone else measured. It says nothing about any specific site.

- I did not describe 60,000 mg/L as a universal truth. It is a modelled ceiling inside a published 15,000–80,000 range.
- I did not claim the automated tests prove real-world correctness. They prove the system is internally consistent and fails safe.

The hardest follow-up, and my answer

"Your model agrees with your model. Why should I believe any of it?"

Because the check uses numbers my engine never reads. CPCB measured the inlet. Operators reported the reject band. I feed in the first and compare against the second, and nothing in between was adjusted to make them meet. If the salt-mass formulation were wrong, that test would fail — and it is a test, so it fails the build, not just a slide.